THE JORDAN-HOLDER THEOREM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the closing sentence 'We conclude by induction.' follows the exhibited two filtrations of the quotient object and the induction hypothesis stated for them, so it closes the argument rather than deferring a step.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Objects of finite length. Same chapter, Lemma lemma-finite-length. Supplies the equivalent conditions holding for the object of the statement, in particular the existence of filtrations with simple successive quotients.
- Successive quotients of a filtration. Same chapter, the description of the quotients A_i/A_{i-1} and of the quotients induced on a quotient object. Supplies the identifications of the successive quotients used in the induction step.

Source reference. Original lemma 0FCJ: sources/stacks-homology/pinned-original-source.tex, source lines 2229–2229. Navigation pointer invoked at proof line 2263; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1 (Jordan-Hölder)**.** *Let $\mathcal{A}$ be an abelian category. Let A be an object of $\mathcal{A}$ satisfying the equivalent conditions of Lemma 0FCJ. Given two filtrations*

$$0 \subset A_1 \subset A_2 \subset \ldots \subset A_n = A \quad \text{and} \quad 0 \subset B_1 \subset B_2 \subset \ldots \subset B_m = A$$

with $S_i = A_i/A_{i-1}$ and $T_j = B_j/B_{j-1}$ simple objects we have $n = m$ and there exists a permutation σ of $\{1, \ldots, n\}$ such that $S_i \cong T_{\sigma(i)}$ for all $i \in \{1, \ldots, n\}$.

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Proof. Let j be the smallest index such that $A_1 \subset B_j$. Then the map $S_1 = A_1 \to B_j/B_{j-1} = T_j$ is an isomorphism. Moreover, the object $A/A_1 = A_n/A_1 = B_m/A_1$ has the two filtrations

$$0 \subset A_2/A_1 \subset A_3/A_1 \subset \ldots \subset A_n/A_1$$

and

$$0 \subset (B_1 + A_1)/A_1 \subset \ldots \subset (B_{j-1} + A_1)/A_1 = B_j/A_1 \subset B_{j+1}/A_1 \subset \ldots \subset B_m/A_1$$

We conclude by induction. □